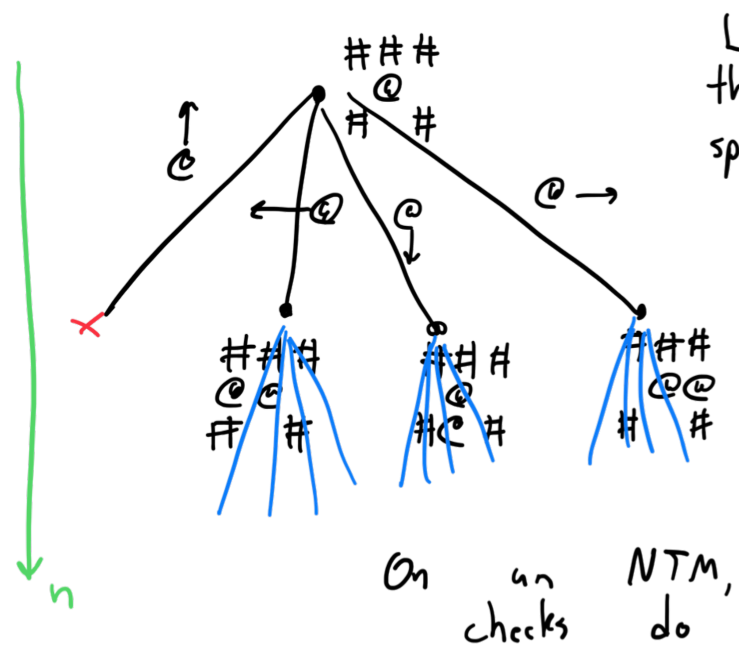
Theorem: A nondeterministic TM (NTM) and a deterministic TM have equivalent expressive power.

Proof: Let N be a NTM. We will construct an equivalent standard TM S. We encode all possible configurations of N (the current state, tape memory, and tape head) on S's tape. We simulate the computation tree of N on S. We perform Depth-first-search on this tree to explore all of the computation of N. □

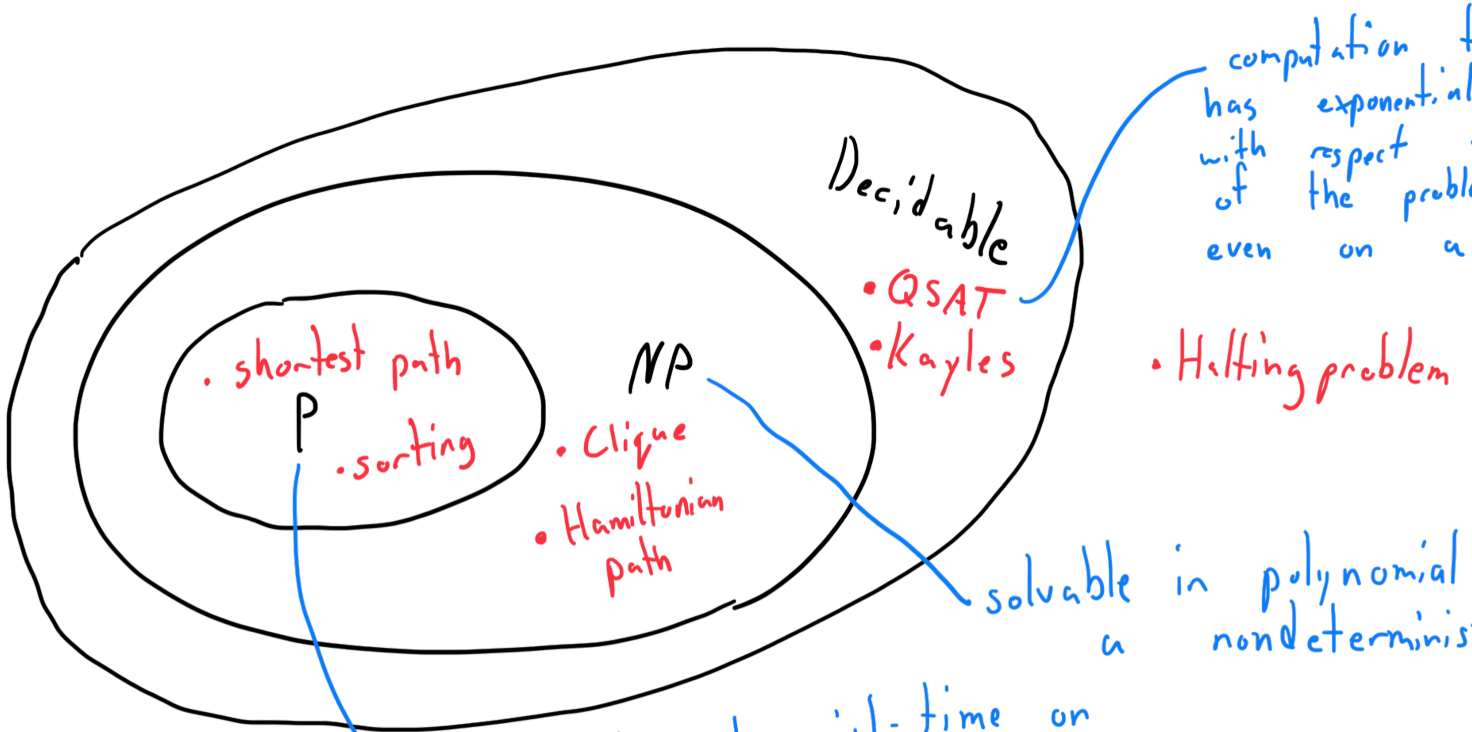
Solving a maze



Let n be the number of spaces in the board.

Worst case # of states (configurations) to check is $O(4^n)$.

On an NTM, how many checks do I make? $O(n)$

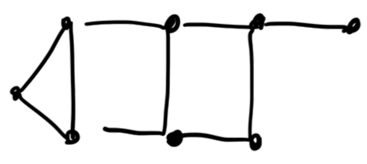


computation tree has exponential depth with respect to the size of the problem state space even on a NTM.

solvable in polynomial-time on a nondeterministic TM.

solvable in polynomial-time on a deterministic TM.

Problem: Does a given graph G have a Hamiltonian path (HP).
(a path that visits each vertex once without repeating.)



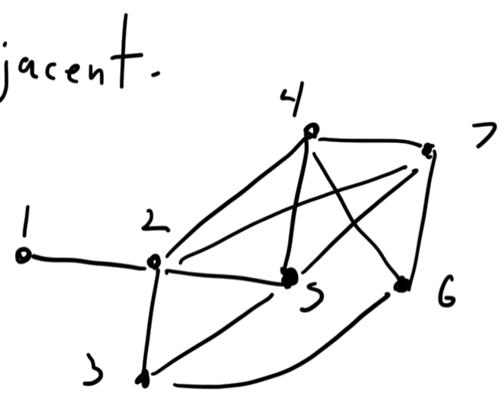
Theorem: HP is decidable.

Proof: Let M = "On input $\langle G \rangle$,

1. Choose a starting vertex we have not chosen before – mark it. If none remain, reject.
2. Randomly mark an adjacent unmarked vertex, making it current. If no such vertex exists, check if all vertices are marked. If so, accept. If not, unmark the current vertex and set current to the previously marked vertex. If no previous marked vertex exists, go to step 1.

(we reserve the far right of the tape as a stack to keep track of the history of marked vertices).

- A clique is a subgraph in a given graph whose vertices are all adjacent.



$Q = \{\langle G, k \rangle \mid$ Graph G has a clique of size $k\}$.

Theorem: Q is decidable

Proof: We iterate through all subsets of the vertices of G up to size k. There are $P(S)$ such sets, where $S \subseteq G$ and $|S| = k$. This can be done with k nested loops.

For each subset, we check if the subset is fully adjacent. If a clique of size k exists, accept. If we exhaust all subsets, reject. □

exponential number of subsets to check!